DIMENSION

Finite and infinite dimension, side by side

2
COMPARED

The finite case is clean and complete; the infinite case needs other tools.

	FINITE-DIMENSIONAL	INFINITE-DIMENSIONAL
DEFINITION	has a finite basis	has no finite basis
BASES	every independent set extends to a basis; every spanning set reduces to one	exist (with the axiom of choice) as Hamel bases, often unwieldy
DIMENSION	a finite integer that governs the whole structure	not a finite number: a Hamel basis for $C[0,1]$ is uncountable
WORKING TOOLS	the theory on this site	Schauder bases, Hilbert spaces, functional analysis
EXAMPLES	$\mathbb{R}^n$, $\mathcal{P}_n$, $\mathbb{R}^{m\times n}$, solution spaces of linear ODEs with constant coefficients	$C[0,1]$; the space of all real sequences

On this site every vector space is finite-dimensional unless stated otherwise.